

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 15

Algorithmic Complexity and the Search for Generative Equations

Parallel Reading

Self-Study Curriculum

Parallel Reading: Ming Li & Paul Vitanyi, *An Introduction to Kolmogorov Complexity and Its Applications* (Springer, 4th ed.), Chapter 2 (“Algorithmic Complexity”), sections 2.1-2.2, the invariance theorem and incompressibility, using Chapter 1 (“Preliminaries”) as a reference to consult when notation blocks you, not as a read-through (core ~50 pages). The book is technical; these sections are its accessible spine.

If unavailable, substitute: MacKay, Chapters 4-6 (source coding, symbol codes, and stream codes) for a more applied treatment.

Also read (short paper): Marcus Hutter, “On Universal Prediction and Bayesian Confirmation” (2007, 15 pages): this connects compression to prediction formally.

Also read: PCM, sections V.20 (“The Insolubility of the Halting Problem”) and V.24 (“The P versus NP Problem”): short entries that pin down the two walls this chapter leans on.